

```
In [28]: pip install --upgrade seaborn
```

```
Requirement already satisfied: seaborn in c:\users\umari\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\umari\anaconda3\lib\site-packages (from seaborn) (1.26.4)
Requirement already satisfied: pandas>=1.2 in c:\users\umari\anaconda3\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\umari\anaconda3\lib\site-packages (from seaborn) (3.10.5)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.59.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\umari\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\umari\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\umari\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\umari\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

```
In [29]: import warnings
warnings.filterwarnings("ignore",category=FutureWarning)
```

```
In [30]: import seaborn as sns
sns.get_dataset_names() #internal inbuilt dataset
```

```
Out[30]: ['anagrams',
'anscombe',
'attention',
'brain_networks',
'car_crashes',
'diamonds',
'dots',
'dowjones',
'exercise',
'flights',
'fmri',
'geyser',
'glue',
'healthexp',
'iris',
'mpg',
'penguins',
'planets',
'seaice',
'taxis',
'tips',
'titanic']
```

```
In [31]: import seaborn as sns
import matplotlib.pyplot as plt

# Load sample dataset
tips=sns.load_dataset('tips')
```

```
# set visual theme
sns.set_theme(style='darkgrid')
```

In [32]:

```
tips
```

Out[32]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

In [33]:

```
tips.to_csv("tips_dataset.csv",index=False)
```

In [34]:

```
import os
os.getcwd()
```

Out[34]:

```
'c:\\Users\\umari\\OneDrive\\Desktop\\Data Science And Ai'
```

In [35]:

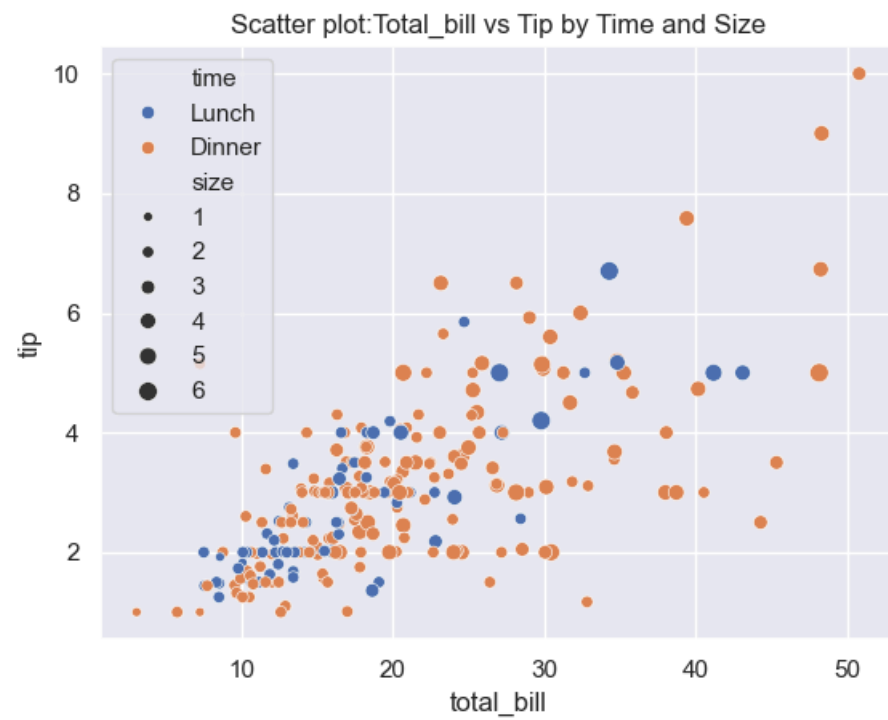
```
plt.figure(figsize=(8,6)) # 8 is width and 6 is height
```

Out[35]:

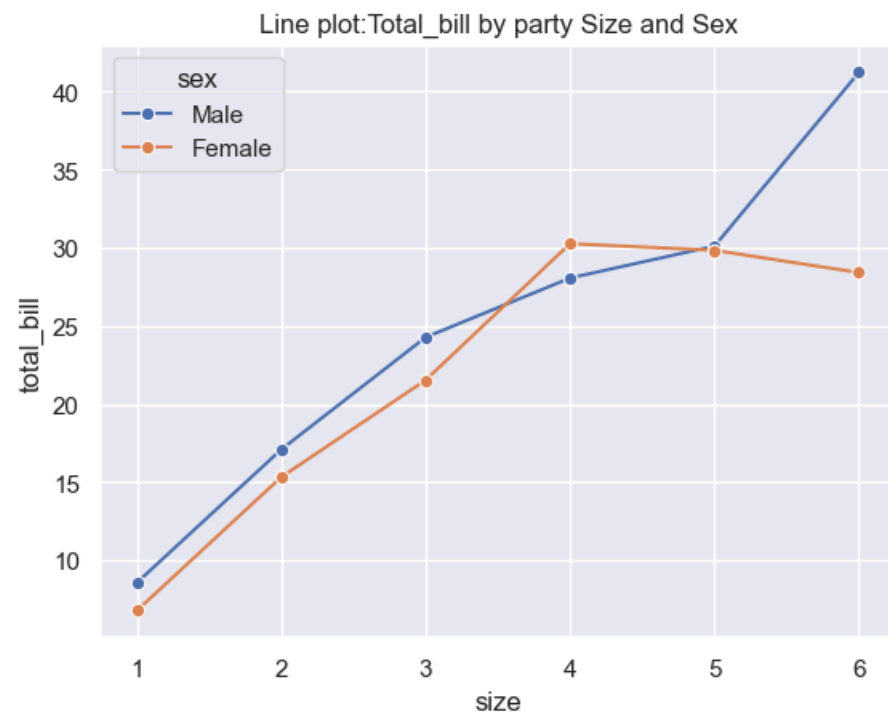
```
<Figure size 800x600 with 0 Axes>
<Figure size 800x600 with 0 Axes>
```

In [36]:

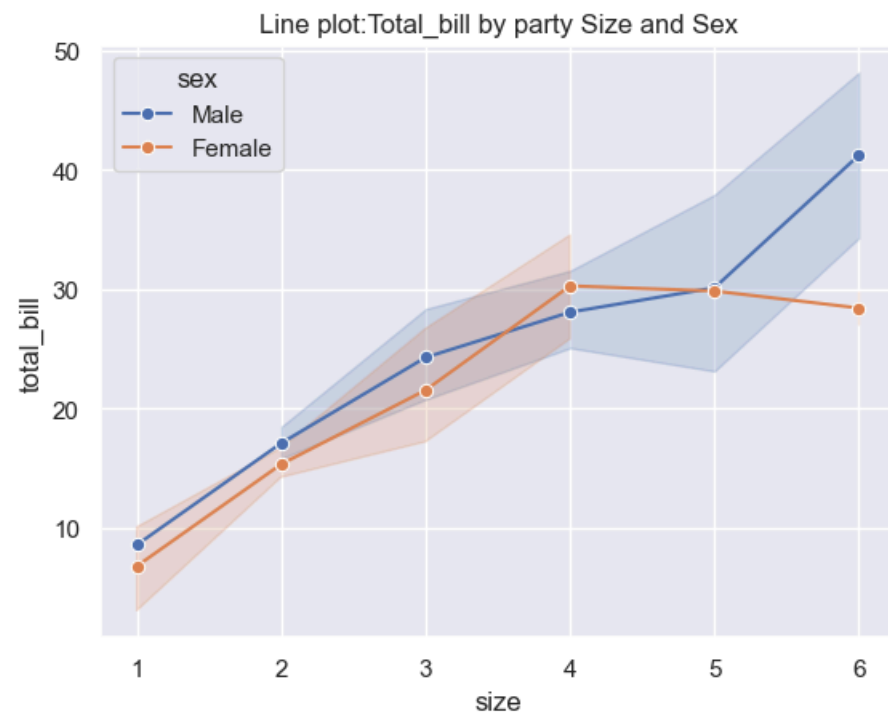
```
# 1.Scatter Plot: Total Bill vs Tip
sns.scatterplot(data=tips,x="total_bill",y="tip",hue="time",size="size",palette="deep")
plt.title("Scatter plot:Total_bill vs Tip by Time and Size")
plt.show()
```



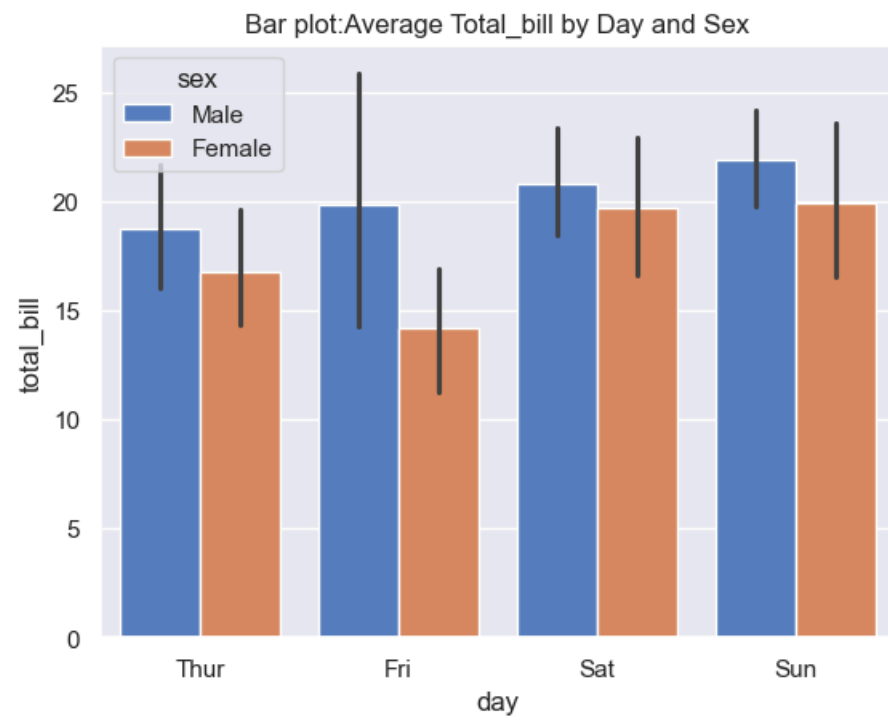
```
In [37]: # 2. Line Plot: Total Bill by Party Size, split by sex
sns.lineplot(data=tips, x="size", y="total_bill", hue="sex", ci=None, marker="o")
plt.title("Line plot: Total_bill by party Size and Sex")
plt.show()
```



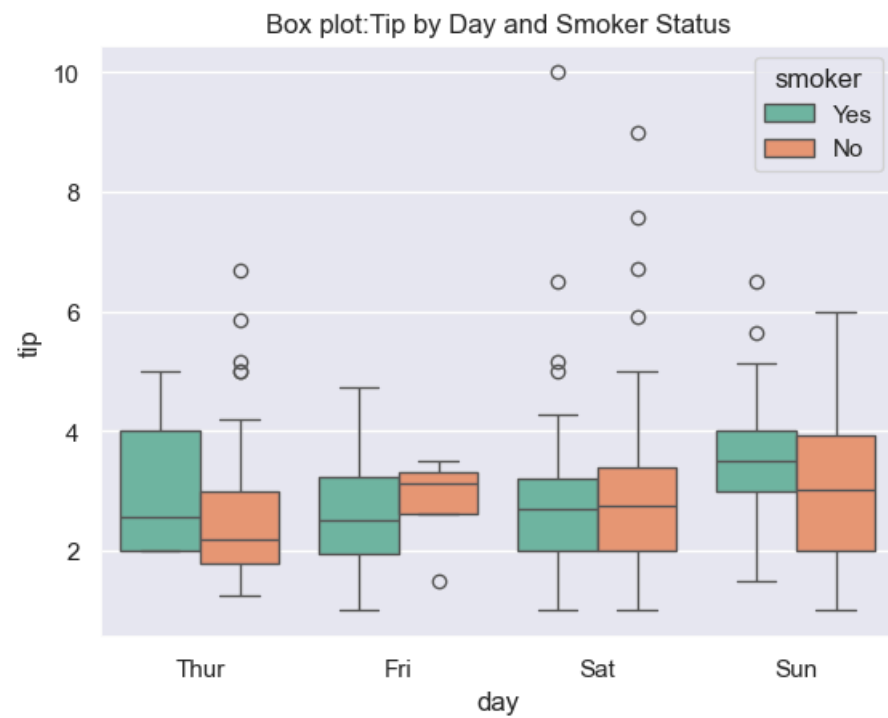
```
In [38]: # Line Plot
sns.lineplot(data=tips,x="size",y="total_bill",hue="sex",ci=95,marker="o")
plt.title("Line plot:Total_bill by party Size and Sex")
plt.show()
```



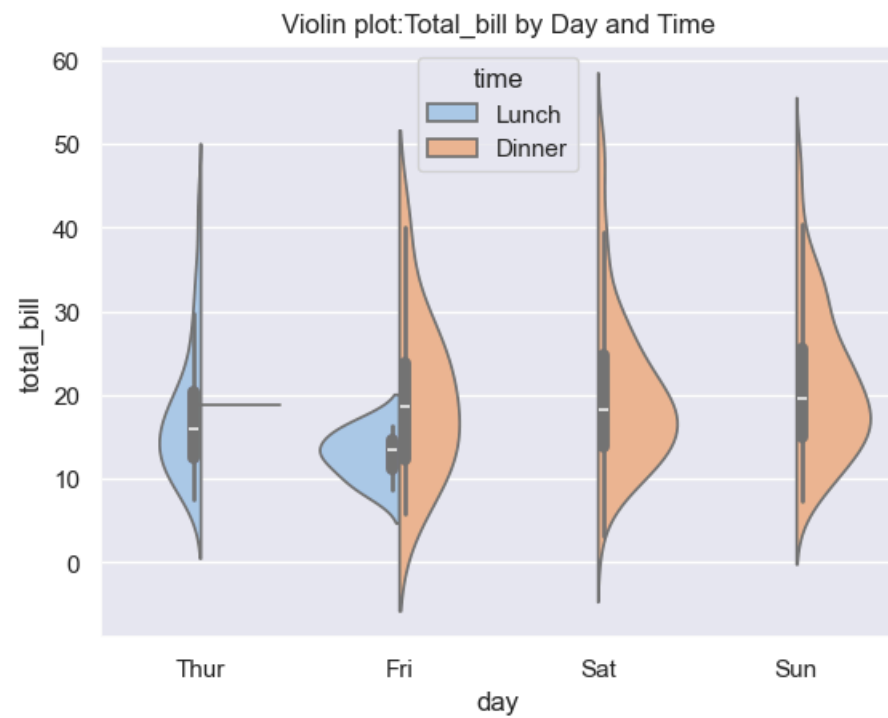
```
In [39]: # Bar Plot: Average Total_bill by Day and split by Sex
sns.barplot(data=tips,x="day",y="total_bill",hue="sex",palette="muted") #muted-less colorful
plt.title("Bar plot:Average Total_bill by Day and Sex")
plt.show()
```



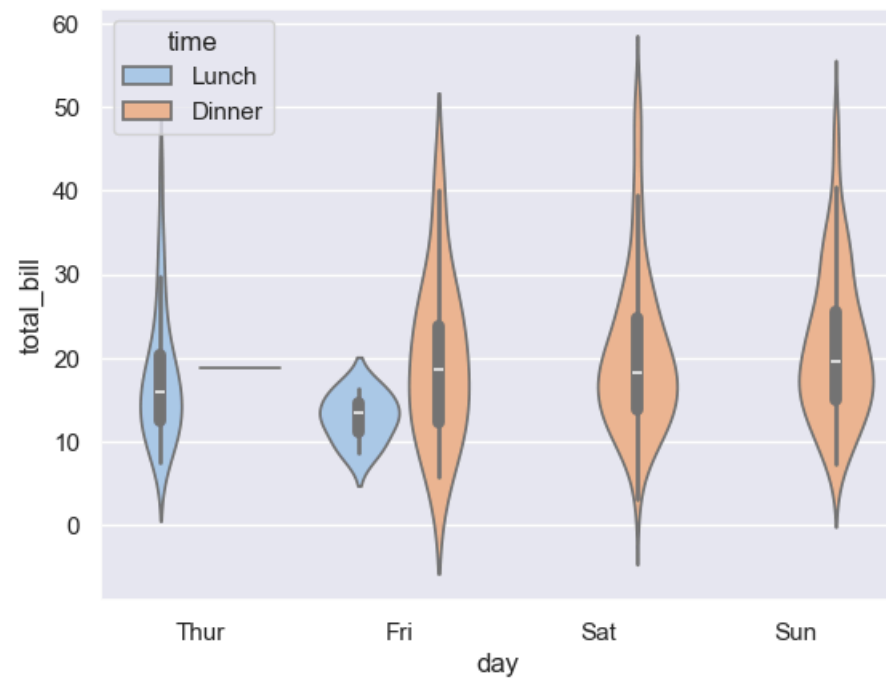
```
In [40]: # Box plot: Total Bill by day, split by time
sns.boxplot(data=tips, x="day", y="tip", hue="smoker", palette="Set2")
plt.title("Box plot: Tip by Day and Smoker Status")
plt.show()
```



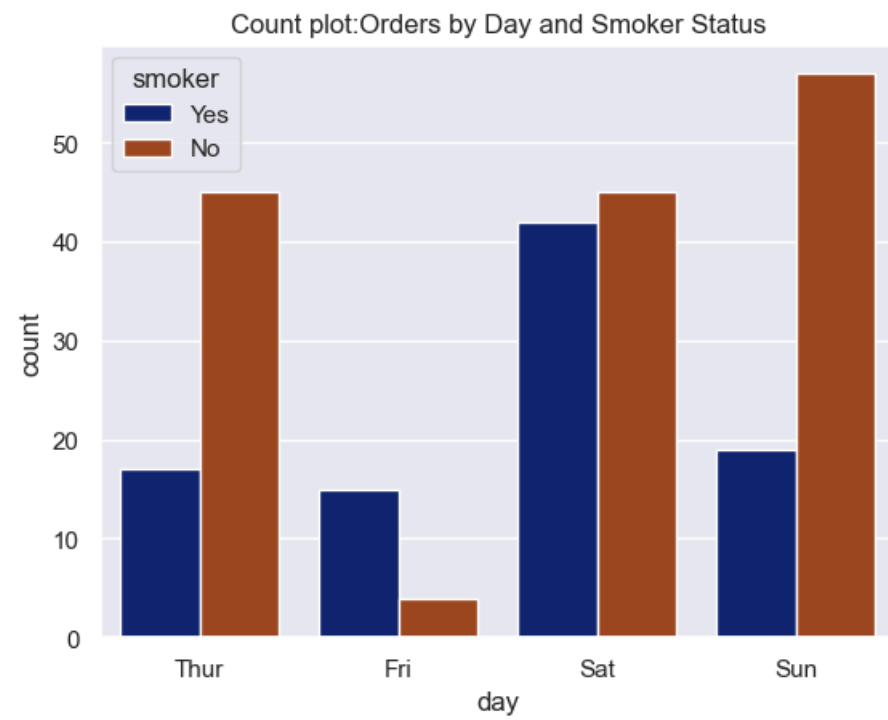
```
In [41]: #Violin Graph/Plot: Total Bill by day,split by time
sns.violinplot(data=tips,x="day",y="total_bill",hue="time",split=True,palette="pastel")
plt.title("Violin plot:Total_bill by Day and Time")
plt.show()
```



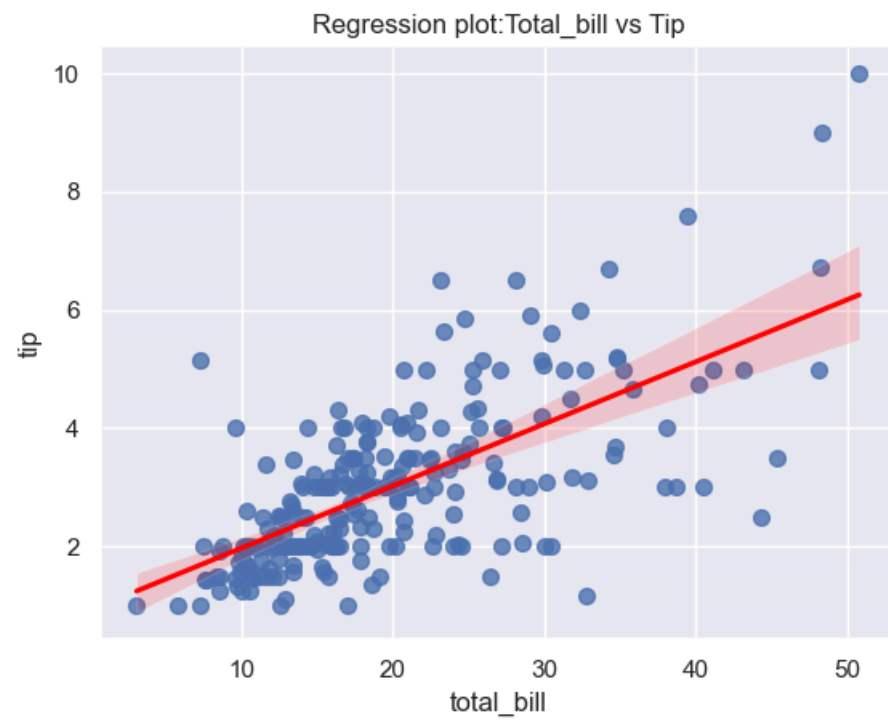
```
In [42]: #Violin Graph/Plot
sns.violinplot(data=tips,x="day",y="total_bill",hue="time",split=False,palette="pastel")
plt.show()
```

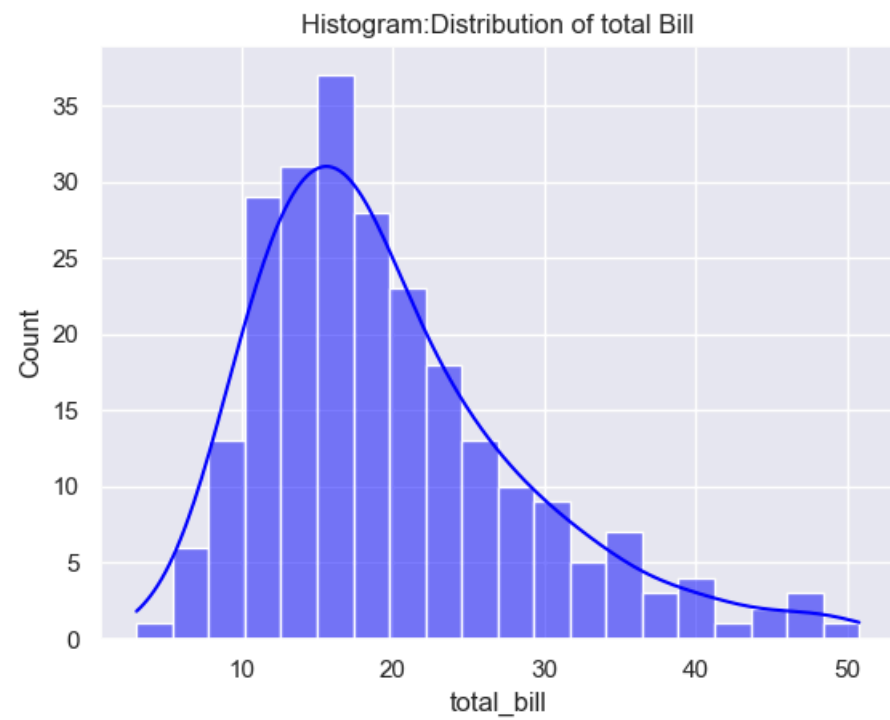
```
In [43]: # Count Plot
sns.countplot(data=tips,x="day",hue="smoker",palette="dark")
plt.title("Count plot:Orders by Day and Smoker Status")
plt.show()
```



```
In [44]: #Regression plot
sns.regplot(data=tips,x="total_bill",y="tip",scatter_kws={"s":50},line_kws={"color":"red"})
plt.title("Regression plot:Total_bill vs Tip")
plt.show()
```



```
In [45]: # Histogram (Distribution Plot): Total Bill with KDE
sns.histplot(data=tips, x="total_bill", kde=True, bins=20, color="blue")
plt.title("Histogram: Distribution of total Bill")
plt.show()
```

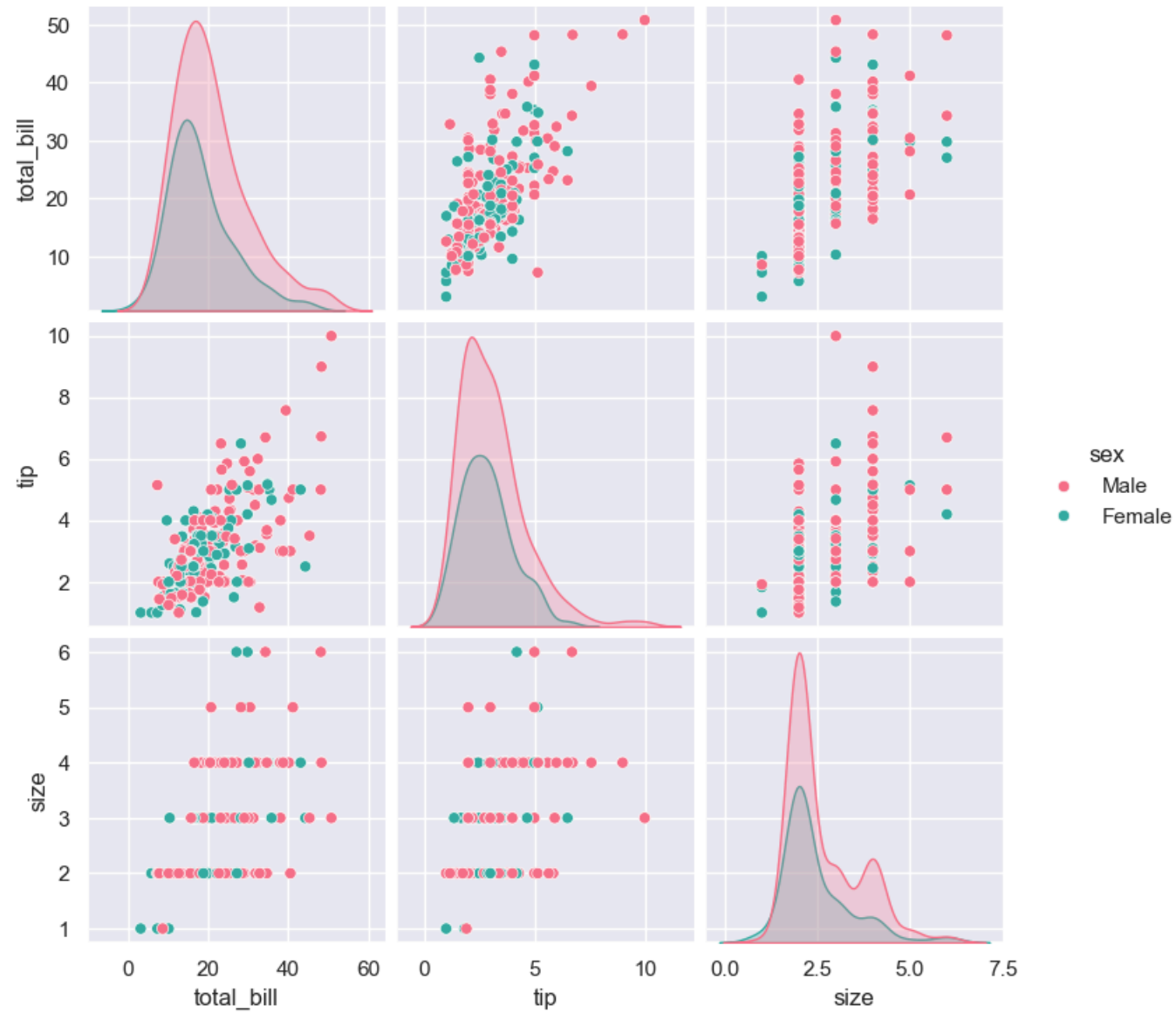


```
In [48]: # Pair Plot: Relationship Between numerical variable
sns.pairplot(tips, hue="sex", vars=["total_bill", "tip", "size"], palette="husl")

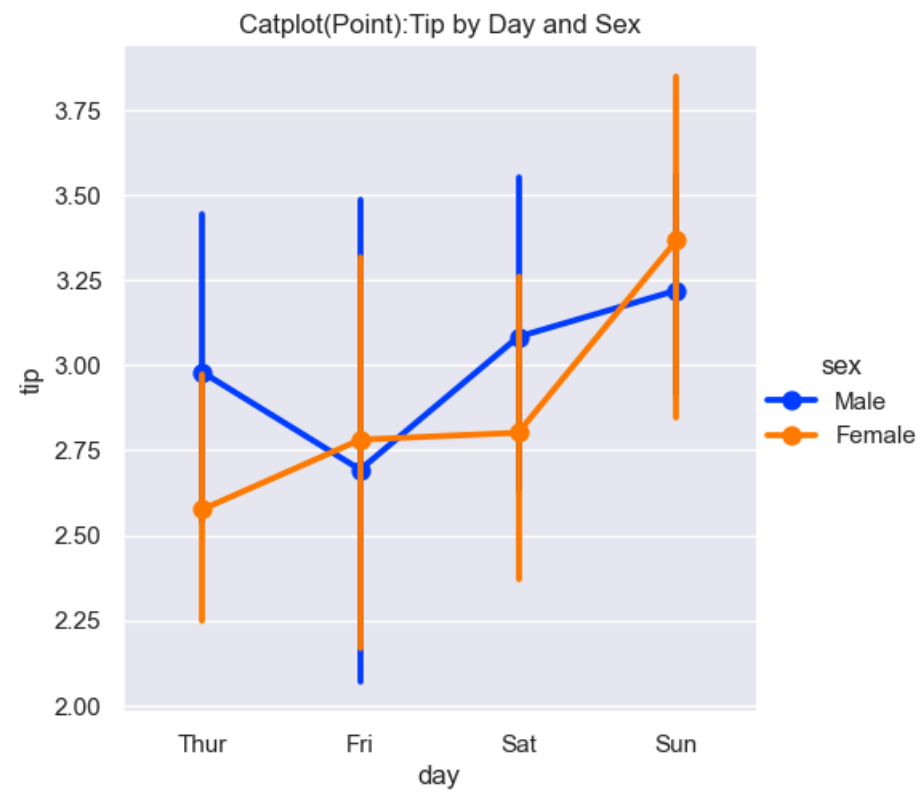
# Correct: suprtile instead of subtitle
plt.suptitle("Pair Plot: Numerical variable by Sex", y=1.02)

plt.show()
```

Pair Plot: Numerical variable by Sex



```
In [49]: # CatPlot(Point Plot): Tip by Day and Sex
sns.catplot(data=tips,x="day",y="tip",hue="sex",kind="point",palette="bright")
plt.title("Catplot(Point):Tip by Day and Sex")
plt.show()
```

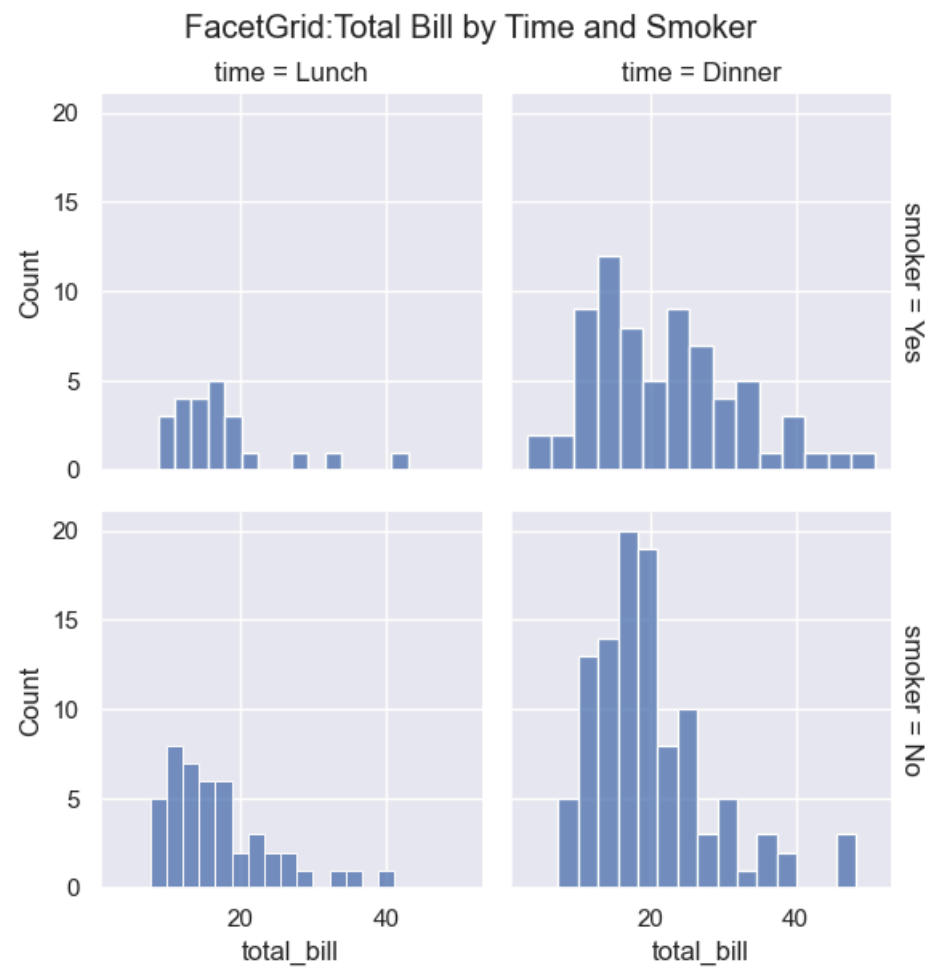


```
In [50]: #Joint Plot:Total Bill vs Tip with marginal Distributions
sns.jointplot(data=tips,x="total_bill",y="tip",kind="scatter",hue="smoker",palette="coolwarm")
plt.suptitle("Joint Plot:Total Bill vs Tip by smoker",y=1.02)
plt.show()
```

Joint Plot: Total Bill vs Tip by smoker



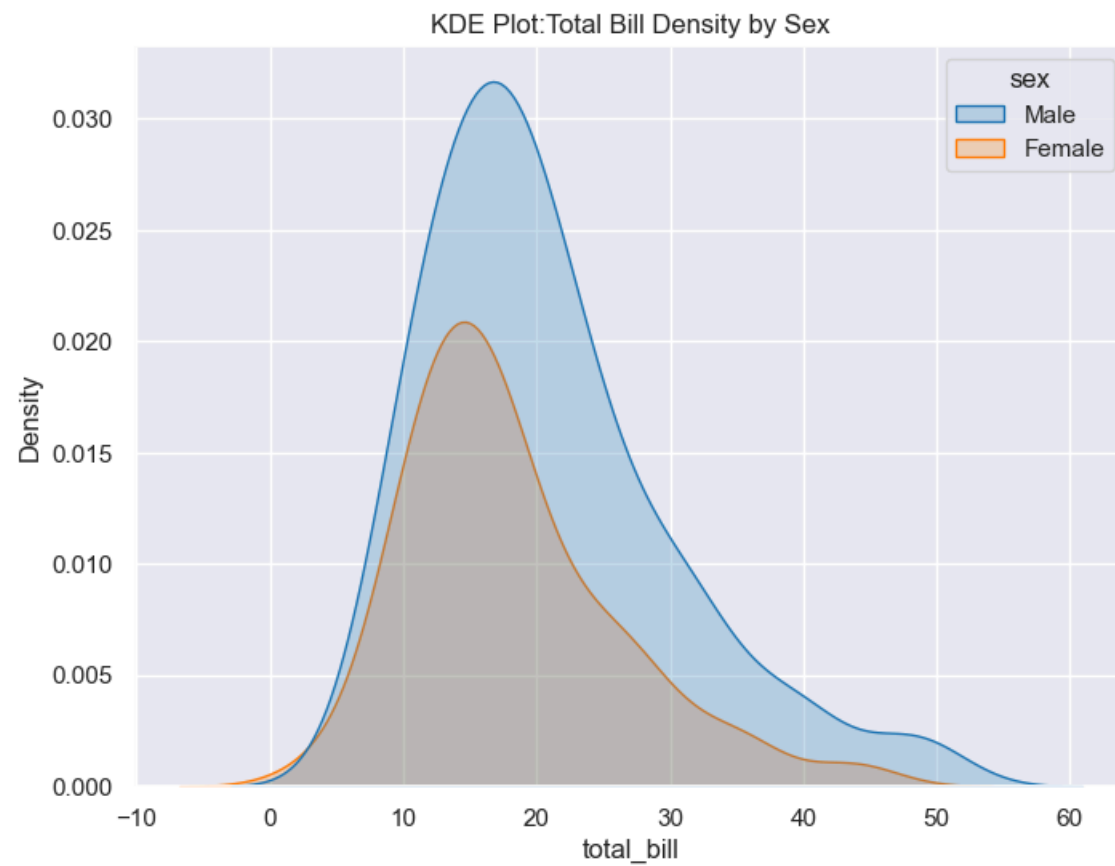
```
In [51]: # FacetGrid: Total Bill by day, Facted by Time and Smoker
g=sns.FacetGrid(tips,col="time",row="smoker",margin_titles=True)
g.map(sns.histplot,"total_bill",bins=15)
g.fig.suptitle("FacetGrid: Total Bill by Time and Smoker",y=1.02)
plt.show()
```



```
In [53]: # Strip Plot: Tips by Day, colored by sex
plt.figure(figsize=(8,6))
sns.stripplot(data=tips, x="day", y="tip", hue="sex", palette="Set1", jitter=True)
plt.title("Strip Plot: Tip by Day and Sex")
plt.show()
```




```
In [54]: # KDE Plot: Total Bill density by Sex
plt.figure(figsize=(8,6))
sns.kdeplot(data=tips, x="total_bill", hue="sex", fill=True, palette="tab10")
plt.title("KDE Plot: Total Bill Density by Sex")
plt.show()
```



Explanations of Plot

-> Scatter Plot:- Shows Relationship Between total_bill and tip, with time and size as additional dimension -> LinePlot:- Trends of total_bill by day, split by sex -> BarPlot:- Average total_bill by day, split by sex -> BoxPlot:- Distribution of tip by day, split by smoker -> Violin Plot:- Distribution of total bill by day, split by time -> Count Plot:- Counts of Orders by day, split by smoker -> Regression Plot:- Linear fit between total bill and tip -> Histogram:- Distribution of total_bill with a KDE curve -> Pair Plot:- PairWise relationship between total bill, tip, and size, colour by sex